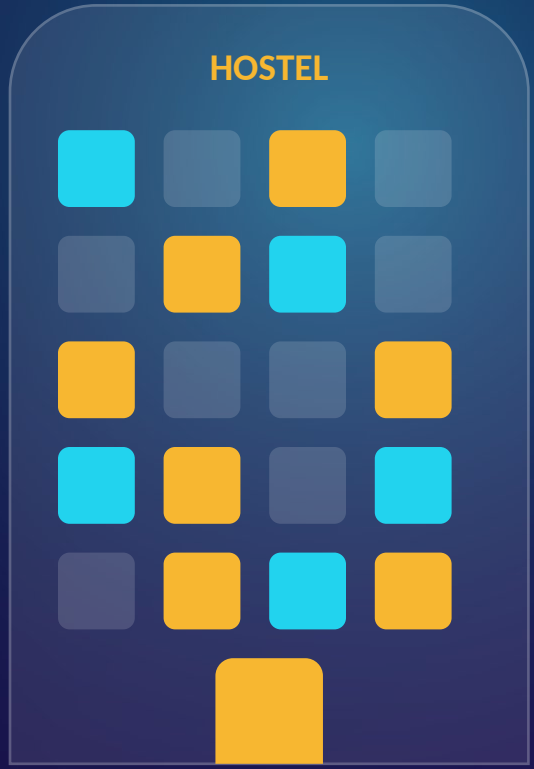
BIRLA VISHVAKARMA MAHAVIDYALAYA

HostelSense

Smart Hostel Fee & Room Report System

A C++ OOP practical project



PRESENTED BY

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ID No.: 25EC407

Problem Statement



1

Manual calculation

Fees differ by room type, so hand calculation invites mistakes.

2

Duplicate entries

The same student can be registered twice without any check.

3

Unclear bills

Shared AC electricity cost is hard to split and mix with the hostel fee.

HostelSense automates all three and saves a clear report.

Objectives

1. Calculate the hostel fee by room type
2. Block the same roll number from registering twice
3. Split the AC electricity bill fairly
4. Save the report to a text file

Technology Used

Technology	Purpose
C++	Core programming language
Online GDB	Writing, compiling and running the code
iostream	Console input and output
fstream	Saving the report to a file
string	Names, room codes and roll numbers

Concepts: classes, inheritance, arrays, loops, conditions, file handling

Main Features

Three room types

Rs. 95,000, Rs. 73,000 and Rs. 1,25,000 per student

Unique roll numbers

College roll no. cannot repeat in a room

AC electricity bill

Units x rate, split equally among 3 students

Separate bills

Electricity is never mixed into the hostel fee

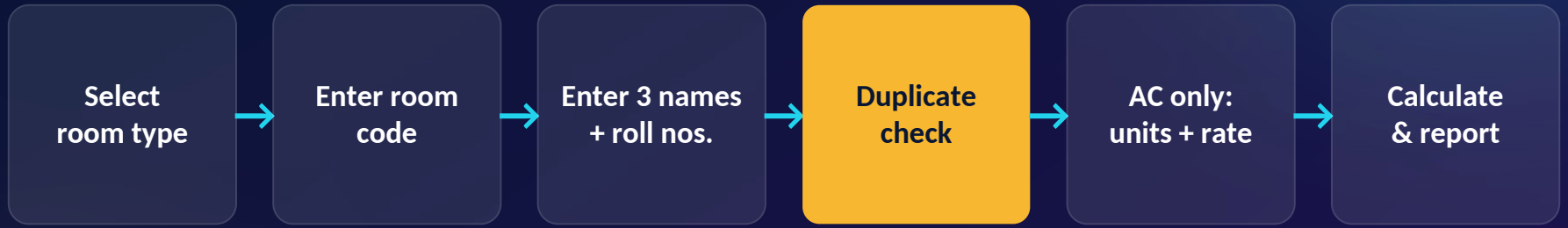
Saved report

Printed on screen and written to HostelReport.txt

Input validation

Invalid room choice or negative values are rejected

How the Program Works



If the roll number already exists, the program asks again until a new one is entered.

Output goes to the screen and to HostelReport.txt.

OOP Concepts Used

1	**Class & objects**	Room holds room code, names and roll numbers
2	**Inheritance**	ACRoom extends Room and adds units, rate, bill()
3	**Encapsulation**	Data and the functions using it live together
4	**Function reuse**	One show(ostream&) prints to screen and file
5	**File handling**	ofstream writes HostelReport.txt

Code: Unique Roll Number

```
while (true) {
    cout << "Enter college roll no: ";
    cin >> rollNo[i];

    bool duplicate = false;
    for (int j = 0; j < i; j++)
        if (rollNo[j] == rollNo[i])
            duplicate = true;

    if (duplicate)
        cout << "Roll no already used!";
    else
        break;
}
```

How it works

- Reads a roll number for the student
- Compares it with every earlier student in the room
- If it matches, shows a message and asks again
- Otherwise the loop ends and the roll number is saved

Code Walkthrough

main.cpp

```
string names[3];
string rollNo[3];

void input() {
    cout << "Enter room code (e.g. 4S13): ";
    cin >> roomCode;

    for (int i = 0; i < 3; i++) {
        cout << "Enter student " << i + 1 << " name: ";
        getline(cin >> ws, names[i]);

        // unique college roll number
        while (true) {
            cout << "Enter student " << i + 1 << " college roll no: ";
            cin >> rollNo[i];

            bool duplicate = false;
            for (int j = 0; j < i; j++) {
                if (rollNo[j] == rollNo[i])
                    duplicate = true;
            }

            if (duplicate)
                cout << "This roll no already lives in the hostel! Enter again.\n";
            else
                break;
        }
    }
}

void show(ostream &out, string type, double bill, double fee) {
    double share = bill / 3;

    out.setf(ios::fixed);
    out.precision(2);

    out << "--- HostelSense Report ---\n";
```

Watch it run

- Screen recording of the program in Online GDB
- Click the video to play it in slideshow mode

Program Output

```
--- HostelSense Report ---
Room: 4S13
Type: Attached Bathroom, Non-AC

=== HOSTEL FEE ===
Student 1: A (Roll no: 25EC407)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 2: B (Roll no: 25ME013)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 3: C (Roll no: 25EL607)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Total hostel fee from this room: Rs. 285000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 0.00
A (Roll no: 25EC407) pays: Rs. 0.00
B (Roll no: 25ME013) pays: Rs. 0.00
C (Roll no: 25EL607) pays: Rs. 0.00
```

Non-AC attached room (4S13): electricity bill Rs. 0

```
--- HostelSense Report ---
Room: 3S12
Type: AC with Attached Bathroom

=== HOSTEL FEE ===
Student 1: D (Roll no: 25PE307)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 2: E (Roll no: 24ME300)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 3: F (Roll no: 25EL402)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Total hostel fee from this room: Rs. 375000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 300.00
D (Roll no: 25PE307) pays: Rs. 100.00
E (Roll no: 24ME300) pays: Rs. 100.00
F (Roll no: 25EL402) pays: Rs. 100.00
```

AC room (3S12): Rs. 300 split as Rs. 100 each, kept separate from the Rs. 1,25,000 fee

Conclusion & Future Scope

Conclusion

- Fees are set by room type automatically
- Duplicate roll numbers are blocked
- Electricity is billed separately
- OOP, validation and file handling in one program

Future Scope

- Many rooms in one run, roll check across the whole hostel
- Search a student by roll number
- Fine or damage charges
- Database or GUI version

Thank You

Any Questions?

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